Name: _____ Date: _____ Period: _____

Integrity Statement:

I acknowledge that I will complete this practice assessment using only my own work. Show all reasoning for free-response questions.

Unit 3 Quiz 1 - Forces and Dynamics

Mock Exam A · Form 2: Foundations

Multiple Choice Questions: Select the best answer. (1 point each)

1. A 40 kg student stands on a scale in an elevator that is moving up and accelerating up, with acceleration magnitude 1 m/s². What force does the scale read? Use g = 9.8 m/s².

(A) 864 N

(B) 216 N

(C) 432 N

(D) -432 N

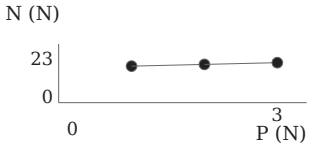
2. Press vertically down on a 2 kg block on a level floor. Use g=10 m/s². P (N) -> N (N): (1, 21); (2, 22); (3, 23). As P (N) increases, what happens to N (N)?

(A) no change

(B) decrease

(C) increase

(D) reversal midway



3. Two masses m1 = 5 kg and m2 = 2 kg hang on an ideal rope over an ideal pulley. m1 moves downward. Find the acceleration magnitude. Use g = 9.8 m/s².

(A) 4.2 m/s²

(B) -4.2 m/s²

(C) 8.4 m/s²

(D) 2.1 m/s²

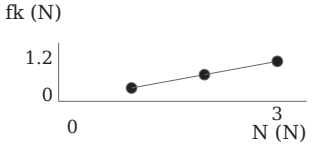
4. The same two surfaces slide against each other. Vary the normal force and measure friction magnitude. N (N) -> fk (N): (1, 0.4); (2, 0.8); (3, 1.2). What is the mean of the first two vertical readings?

(A) 0.36

(B) 0.84

(C) 1.08

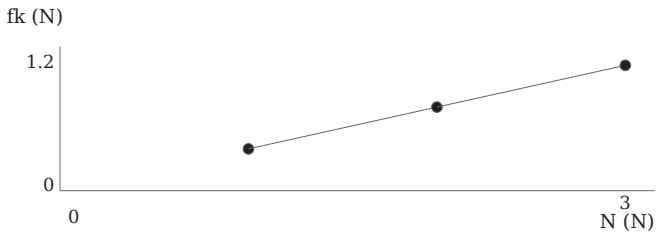
(D) 0.6



Free Response Questions: Answer all parts. Show your work and clearly indicate your answers.

5. The same two surfaces slide against each other. Vary the normal force and measure friction magnitude. N (N) -> fk (N): (1, 0.4); (2, 0.8); (3, 1.2).

a. For the sliding-friction measurements setup, label the measured quantities and their reference directions or origins. **(1 pt)**



b. Plot the first and last measurements and describe the trend. **(3 pt)**

6. Touching 2 kg and 3 kg carts accelerate on a frictionless floor when a hand pushes the 2 kg cart right. Fhand (N) -> Fcontact (N): (1, 0.6); (2, 1.2); (3, 1.8).

a. Build the model for two carts in contact from physics and predict both slope and intercept. **(2 pt)**

b. Plot the first and last measurements and describe the trend. **(2 pt)**

7. Press vertically down on a 2 kg block on a level floor. Use g=10 m/s². P (N) -> N (N): (1, 21); (2, 22); (3, 23). Build the model for downward load from physics and predict both slope and intercept. **(3 pt)**